In [2]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Generate sample website traffic data
np.random.seed(10)

dates = pd.date_range(start='2018-09-01', periods=365, freq='D')

page_views = np.random.randint(20, 100, 365)

# Add traffic spikes
for day in [30, 60, 90, 150, 180, 210, 240, 270, 300, 330, 350]:
    page_views[day] += np.random.randint(100, 1000)

data = pd.DataFrame({
    'Date': dates,
    'Page_Views': page_views,
    'Unique_Visitors': page_views * 0.7,
    'Bounce_Rate': np.random.uniform(20, 80, 365)
})

data.set_index('Date', inplace=True)

# Plot website traffic (similar to sample output)
data['Page_Views'].plot(figsize=(10, 3))
plt.title('Website Traffic')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.show()

# Time series decomposition
decomposition = seasonal_decompose(
    data['Page_Views'], model='additive', period=7
)
decomposition.plot()
plt.show()

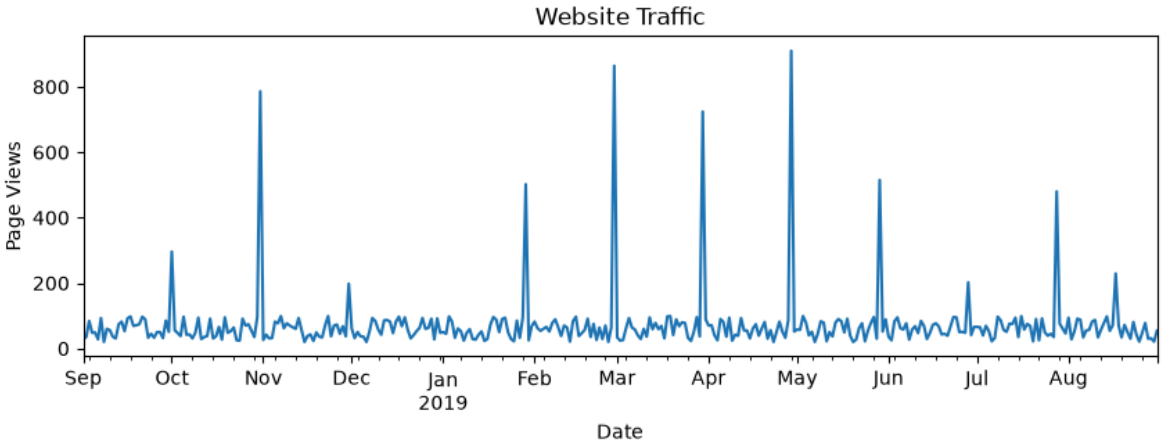
# Moving average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot(figsize=(10, 4))
plt.title('Page Views with Moving Average')
plt.show()

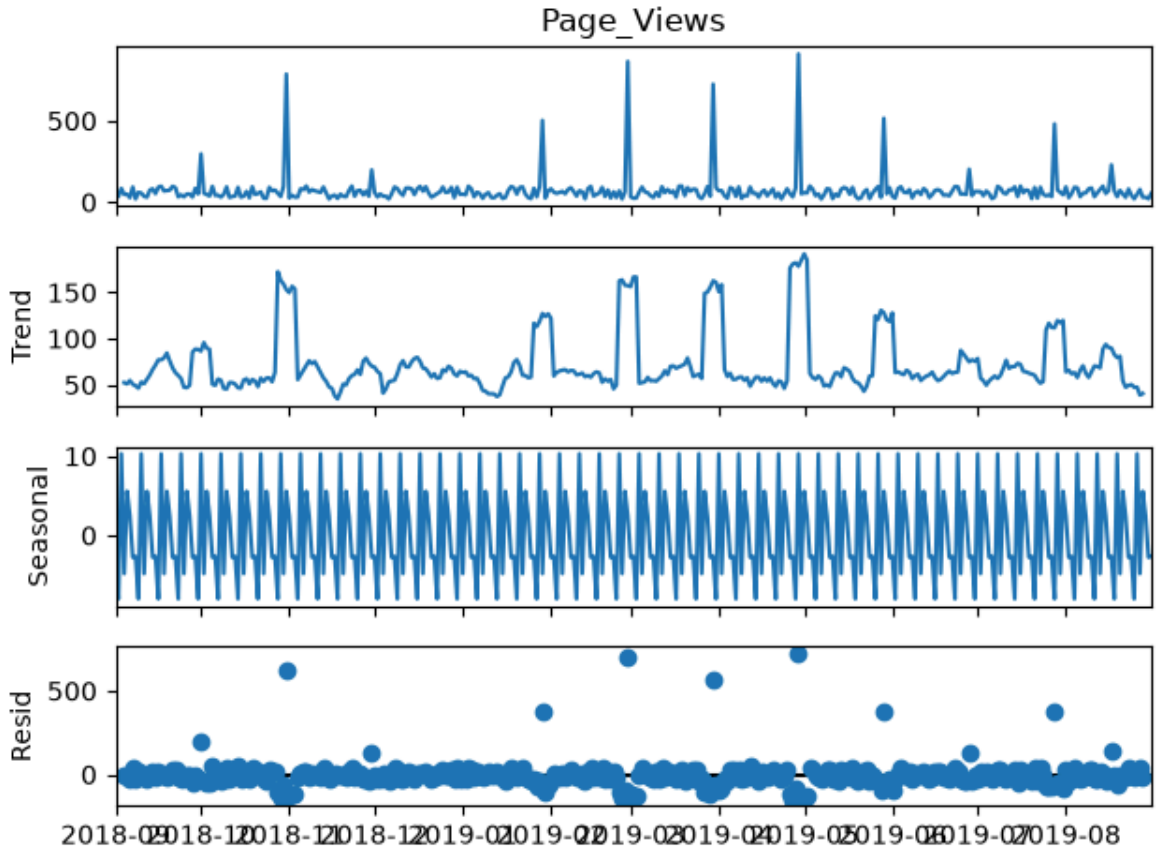
# Plot unique visitors and bounce rate
data[['Unique_Visitors', 'Bounce_Rate']].plot(
    subplots=True, figsize=(10, 5)
)
plt.show()

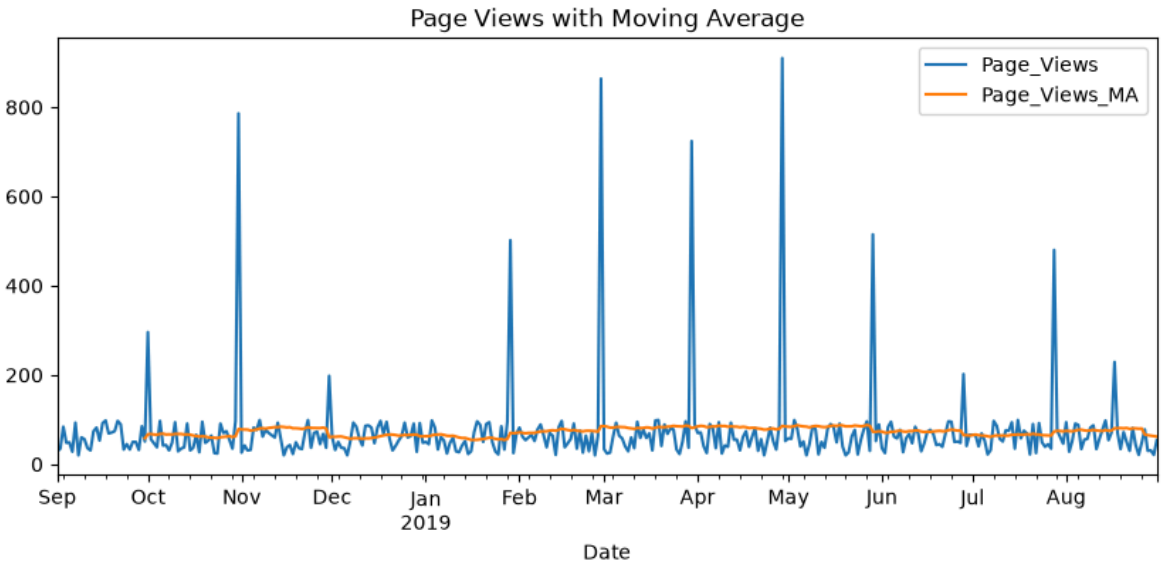
# Anomaly detection using Z-score
data['Page_Views_Z'] = (
    (data['Page_Views'] - data['Page_Views'].mean())
    / data['Page_Views'].std()
)

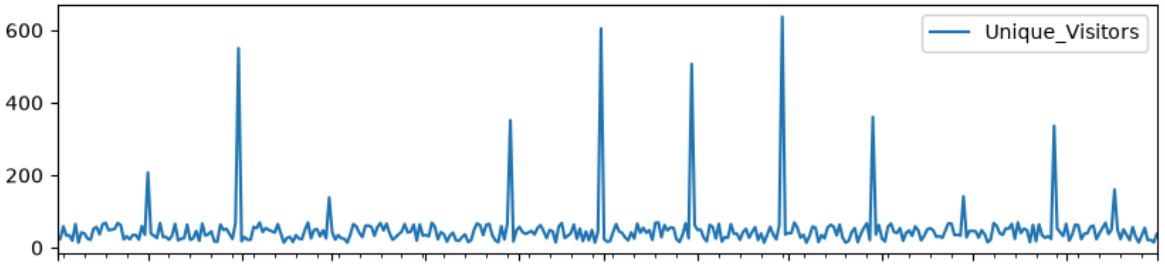
anomalies = data[data['Page_Views_Z'].abs() > 3]
```

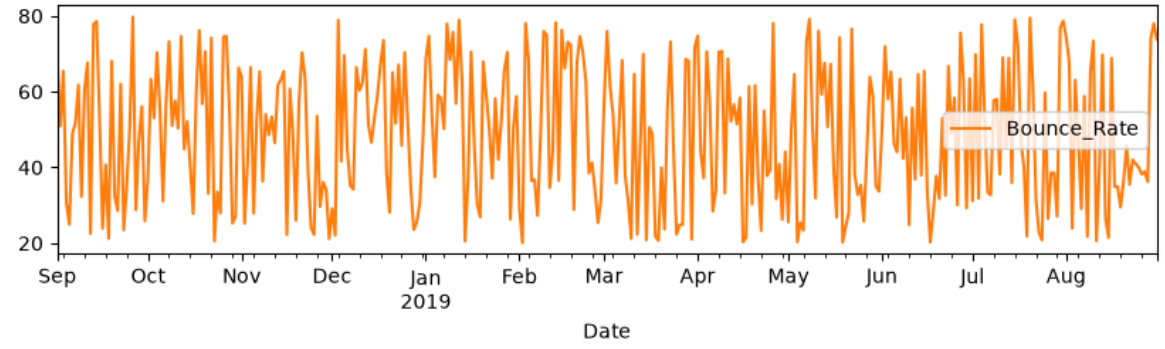
```
plt.figure(figsize=(10, 4))
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)
plt.title('Website Traffic Anomaly Detection')
plt.legend()
plt.show()
```

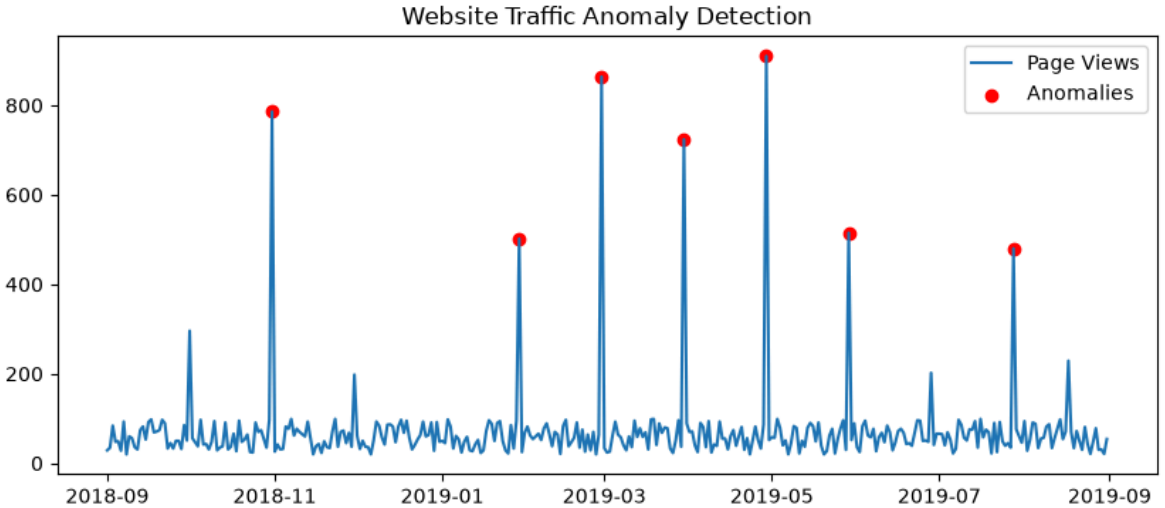





Page Views with Moving Average
Page_Views
Page_Views_MA
800
600
400
200
0
Sep
Oct
Nov
Dec
Jan
2019
Feb
Mar
Apr
May
Jun
Jul
Aug
Date


Unique_Visitors
600
400
200
0


Bounce_Rate
80
60
40
20
Sep
Oct
Nov
Dec
Jan
2019
Feb
Mar
Apr
May
Jun
Jul
Aug
Date


Website Traffic Anomaly Detection
Page Views
Anomalies
800
600
400
200
0
2018-09
2018-11
2019-01
2019-03
2019-05
2019-07
2019-09

In [ ]: